Pascal Distribution

This distribution is also known as the Negative Binomial Distribution.

In this scenario, a Bernoulli trial is repeated until k successes are observed.

$$f(x, k, p) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, x \geq k$$

This distribution models the number of total repetitions before the k^{th} success.

Expected Values

These are the moments for the number of total trials:

$$E[X] = \frac{k}{p}$$

$$Var[X] = \frac{k(1-p)}{p^2}$$

If measuring failures, we have:

$$E[X] = \frac{k(1-p)}{p}$$

$$Var[X] = \frac{k(1-p)}{p^2}$$